

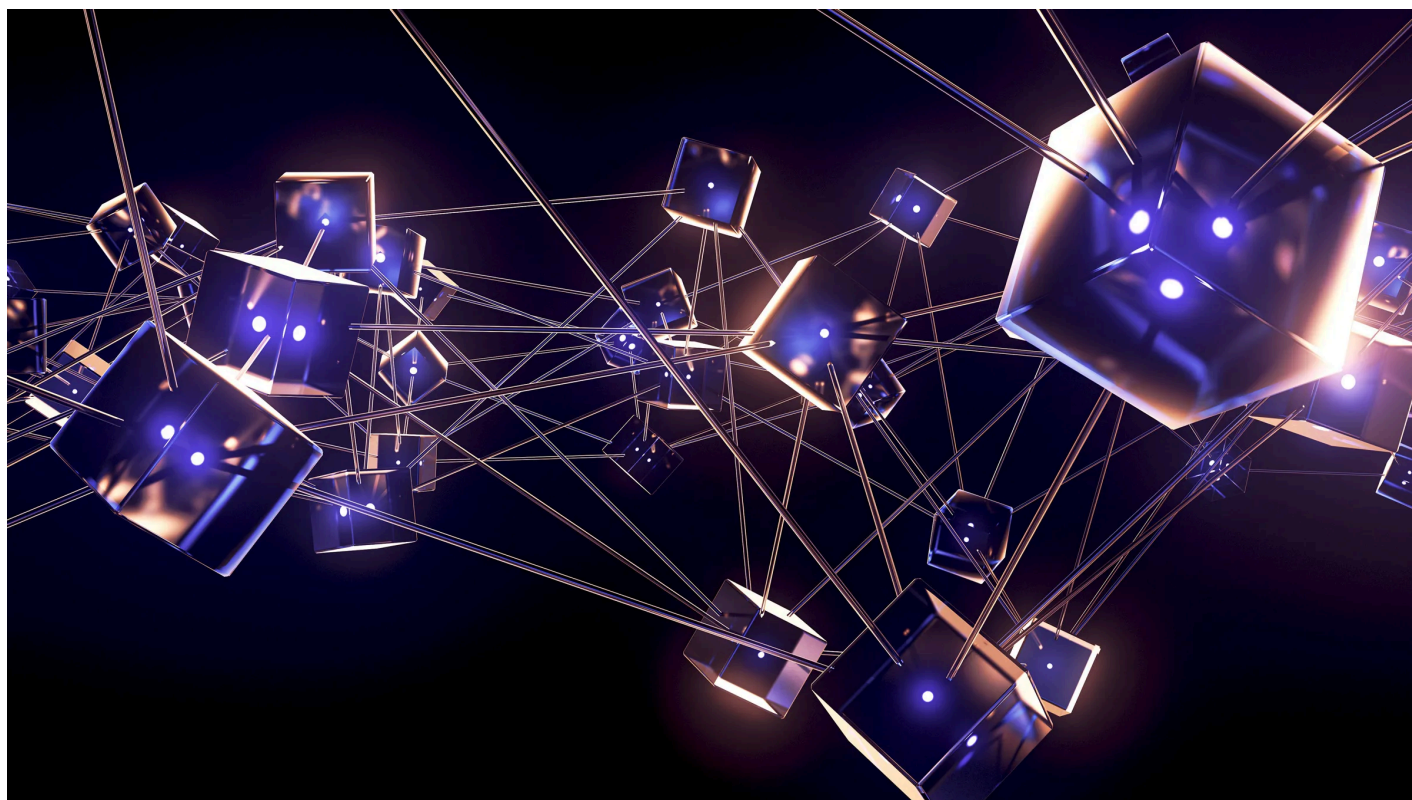


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IEEE Std 3221.01–2025: IEEE Standard for Blockchain Interoperability—Cross Chain Transaction Consistency Protocol



Introduction

With the development of blockchain technology, numerous blockchain protocols and networks have emerged. However, the lack of direct value and information exchange between these networks has created a siloed environment that significantly limits the widespread adoption of blockchain. Due to the absence of effective interoperability between different blockchain networks, frequent reconfiguration and integration efforts are required. In order to address the interoperability concern, the IEEE Computer Society Blockchain and Distributed Ledger Standards Committee developed and published a new standard, IEEE Standard 3221.01–2025 (IEEE

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Standard for Blockchain Interoperability – Cross Chain Transaction Consistency Protocol), that provides a high-performance distributed cross-chain transaction consistency protocol to promote blockchain interoperability, enhance service quality and security, and build a collaborative blockchain ecosystem — ultimately unlocking the full potential of blockchain technology.

Overview of the Standard

IEEE Std 3221.01–2025 establishes a cross-chain consistency framework for multiple cross-chain technologies, including notary-based (centralized/multi-signature), Hash Time-Locked Contract (HTLC)-based, and relay chain-based architectures. It serves as a reference for blockchain project teams, blockchain service providers, decentralized application (dApp) developers, and any entities or individual users seeking to exchange assets and information across different blockchain networks without relying on centralized third parties.

Adoption and Impact

By adopting this standard, practitioners can efficiently design highly collaborative cross-chain systems, thereby accelerating the widespread adoption of blockchain technology and enabling seamless interoperability across heterogeneous blockchain environments.

As an illustrative example, the cross-chain technical framework proposed in this standard has been applied in China's national blockchain innovation application pilot programs. The blockchain infrastructure platform in Jing'an District, Shanghai, China, adopted the standardized cross-chain solution, successfully enabling trusted data transmission across blockchains and achieving interoperability among multiple blockchain applications. This implementation has demonstrated blockchain's value in promoting data sharing, optimizing business processes, and improving collaborative efficiency, thus contributing to high-quality economic and social development.

Future Developments

The working group responsible for this standard has identified three key research directions for future development: Application of quantum-resistant signatures in cross-chain protocols, Lightweight node verification schemes based on zero-knowledge proofs, and Atomic swap specifications for cross-chain NFTs.

Conclusion

IEEE Std 3221.01–2025 is the latest standards from the IEEE Computer Society Blockchain and Distributed Ledger Standards Committee that published cross-chain consistency specifications for notary-based, HTLC-based, and relay chain-based architectures. It provides practical guidance for blockchain project teams, service providers, dApp developers, and users who seek to exchange assets and information across different blockchain networks without centralized intermediaries. The case study shows that this standard significantly improves the security, scalability, and flexibility of cross-chain technologies, and plays a critical role in promoting data sharing, optimizing business processes, and enhancing cross-organizational collaboration.



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